Data Privacy and Security

Anonymization

Understanding the Role of Anonymization in Data Protection

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Outline

- Understanding Anonymization
- Anonymization Techniques
- Differential Privacy – The Gold Standard
- Ethical & Legal Considerations

Learning Objectives

- Define anonymization and understand its role in data privacy.
- Recognize the importance of anonymization for compliance with laws such as GDPR and CCPA.
- Differentiate between anonymization, pseudonymization, and de-identification.
- Understand real-world examples of anonymization and its challenges.

Why Anonymization Matters?

Removing identifying information from (something, such as computer data) so that the original source cannot be known.

- Aim: Prevent re-identification of individuals.
- Includes removing or transforming personal data.
- Often used before data is published or shared.

Why Anonymization Matters?

Why Is It Important?

- Legal Compliance
  - Required by GDPR (General Data Protection Regulation), CCPA (California Consumer Privacy Act), HIPAA (Health Insurance Portability and Accountability Act), etc.
- Security & Privacy
  - Prevents sensitive data exposure in data breaches.
- Data Sharing
  - Allows organizations to share datasets for research and business without violating privacy regulations.
- Trust
  - Ensures user trust in data-driven applications, such as AI and analytics.

Limitations of Traditional Techniques

Early methods under Statistical Disclosure Control:

- Suppression (drop columns like names, addresses, birthdates, etc.)
- Aggregation (e.g., 5 people in this dataset live in this postal code)
- Scrambling/row swapping

BUT:

- They obfuscate data but don't guarantee privacy.
- Vulnerable to modern re-identification **attacks** or information gain.

Limitations of Traditional Techniques

Re-identification Risk & Information Gain

- Cynthia Dwork's work showed:

"Any data release has the potential for re-identification or information gain."

- Even if someone is not in the dataset, attackers can infer facts using *auxiliary information*.

The Anonymization Myth

- Classical anonymization ≠ perfect privacy.
- Lesson: You can't guarantee someone's identity won't be revealed.
- Example: Knowing a person's height + average population → infer individual's presence.

Real-World Privacy Breaches Due to Poor Anonymization

AOL Search Data Leak (2006)

- AOL released search query logs of 650,000 users.
- Names were removed, but search queries revealed identities (e.g., one user searched "how to kill my wife").
- **Lesson** → Even "anonymized" data can lead to re-identification.

Netflix Prize Data Breach (2007)

- Netflix released an anonymized dataset of user movie ratings for a competition.
- Researchers cross-referenced it with IMDB ratings and **re-identified** users.
- **Lesson** → Anonymized data, when combined with external data, can be deanonymized.

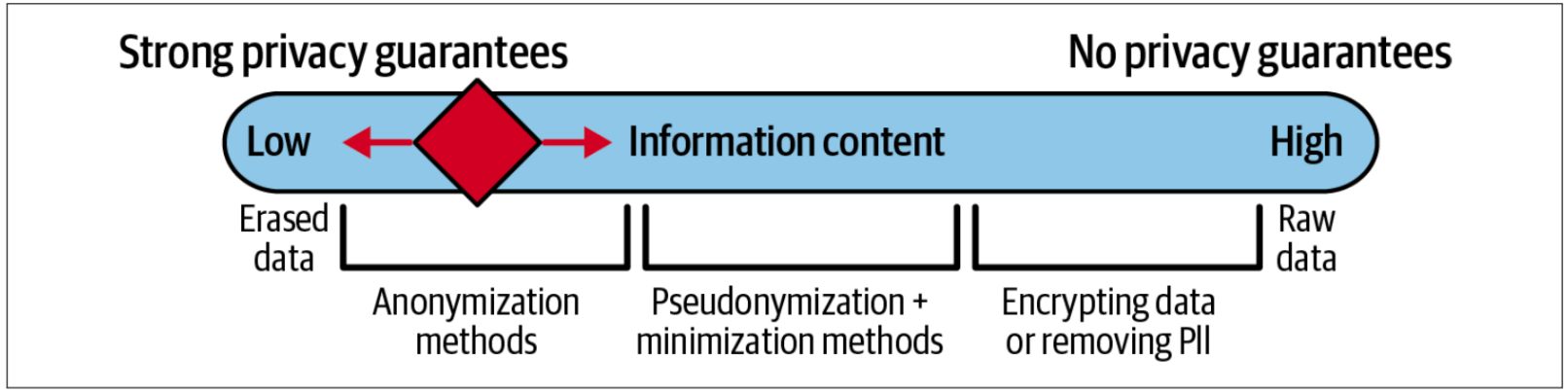
Strava Fitness App Privacy Leak (2018)

- Strava released a heatmap of global jogging routes.
- Military bases were exposed by soldiers' running routes.
- **Lesson** → Even aggregated data can pose security risks.

From Binary to Privacy Continuum

Old mindset: Privacy is either *on* or *off*

New mindset: Privacy exists on a *spectrum*



A continuum from:

Erased data (strong privacy, low utility)
to

Raw data (no privacy, high utility)

Measuring Privacy Loss

- **Privacy loss** = amount of information that can be learned.
- No “perfect anonymization” → instead, quantify risk.
- Key idea: Shift from *anonymize perfectly* to *control how much can be learned*.

Enter Differential Privacy

Introduced to solve the failures of traditional anonymization.

Allows measuring privacy loss (via ε).

- Balances:
  Utility for data analysis
  Guarantees for individual privacy

Defining Differential Privacy

What is Differential Privacy?

- A rigorous definition that limits the privacy loss when releasing data.
- Allows measuring the risk of exposing an individual's information.
- Protects even if the attacker has auxiliary information.]

How Does It Work?

- Adds *random noise* to outputs (using algorithms called mechanisms).
- Noise is calibrated to hide one person's presence or absence.
- Guarantees apply to the process, not just the result.

Defining Differential Privacy

Core Guarantee:

"The output of a query is nearly the same whether or not any one individual is in the dataset."

$$\Pr[A(D_1) \in S] \leq e^{\varepsilon} \cdot \Pr[A(D_2) \in S]$$

Key Terms:

- ε (epsilon): Maximum privacy loss allowed.
- D_1 & D_2: Neighboring datasets (differ by one person).
- Algorithm A: Adds noise, must satisfy privacy bounds.

Goal of Differential Privacy

- Prevent attackers from learning too much about any individual.
- Maintain utility while ensuring privacy is mathematically provable.

What is Epsilon (ε) in Differential Privacy?

- ε controls how much privacy is "spent" per query.
- Each individual's information leakage is bounded by ε.
- Smaller ε = stronger privacy, larger ε = more accurate results.

Privacy Attack Example: Inferring Salary

- Attacker takes a screenshot of salary dashboard before and after you join.
- Compares average change to reverse-engineer your salary.
- Assumes you're the only new employee in that role.

Epsilon as Bounded Information Gain

- Differential Privacy (DP) can be viewed from the attacker's perspective.
- Prior suspicion $D = D_1 \rightarrow$ Updated suspicion (posterior) after query response $A(D) = O$.

$$P[D = D_1 | A(D) = O] = P[D = D_1] \times \frac{P[A(D_1) = O]}{P[A(D) = O]}$$

- The potential information gain based on the epsilon and the attacker's initial suspicion (auxiliary information).

$$\frac{P[D = D_1]}{e^{\varepsilon} + (1 - e^{\varepsilon}) \times P[D = D_1]} \leq P[D = D_1 | A(D) = O] = P[D = D_1] \leq \frac{e^{\varepsilon} \times P[D = D_1]}{1 + (e^{\varepsilon} - 1) \times P[D = D_1]}$$

- Goal: Limit how much suspicion can grow after observing results.

How Epsilon Affects Information Gain

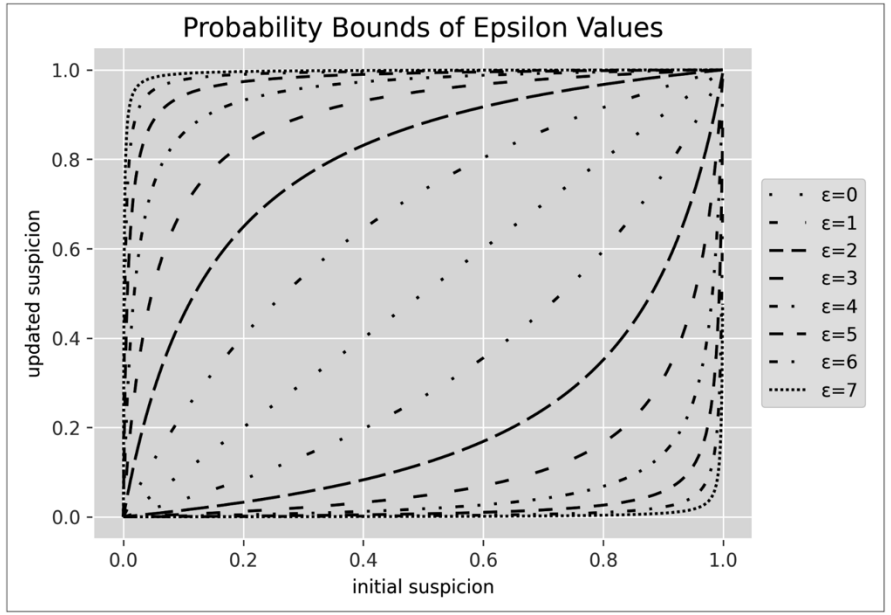


Figure 2-2. Estimating knowledge gain based on epsilon

- Small ε: output gives little new info.
- Larger ε: attacker can become highly confident.

Choosing a Good ε Value

- Small ε (e.g. < 1) keeps privacy leakage low.
- Helps prevent accurate inference—even in worst-case attacks.
- Balances privacy vs utility depending on sensitivity and use case.

Implementing DP: Noise Mechanisms

- Add noise to query results (Laplace, Gaussian).
- Noise is calibrated using ε and query sensitivity.
- Ensures formal privacy guarantees within best/worst-case bounds.

What Differential Privacy Guarantees, and What It Doesn't

Differential privacy is…	Differential privacy is not…
• Quantifies and bounds information gain. • Offers plausible deniability for individuals. • Enables safe data analysis while protecting privacy.	• Not zero-risk or perfect anonymization. • Doesn't block all group-level information. • Not a one-size-fits-all privacy tool.

Plausible Deniability in Action

- DP allows individuals to say: "That result may describe many—but not necessarily me."
- Supports sensitive data research (e.g., gender pay studies).

Anonymization in Data Security: Case Study of the US Census

Why Anonymize Data?

- Protect personal privacy
- Prevent re-identification
- Enable data sharing with minimized risk

The US Census Case

- Conducted every 10 years
- Informs federal funding, representation
- Must count *everyone* exactly once

The Privacy Challenge

- Public access to census data
- Risk: linking with external databases
- 2010 test showed 38% success rate in re-identification

External Threats to Anonymized Data

- Voter records, consumer databases, insurance info
- Used for re-identification
- Also exploited in targeted advertising

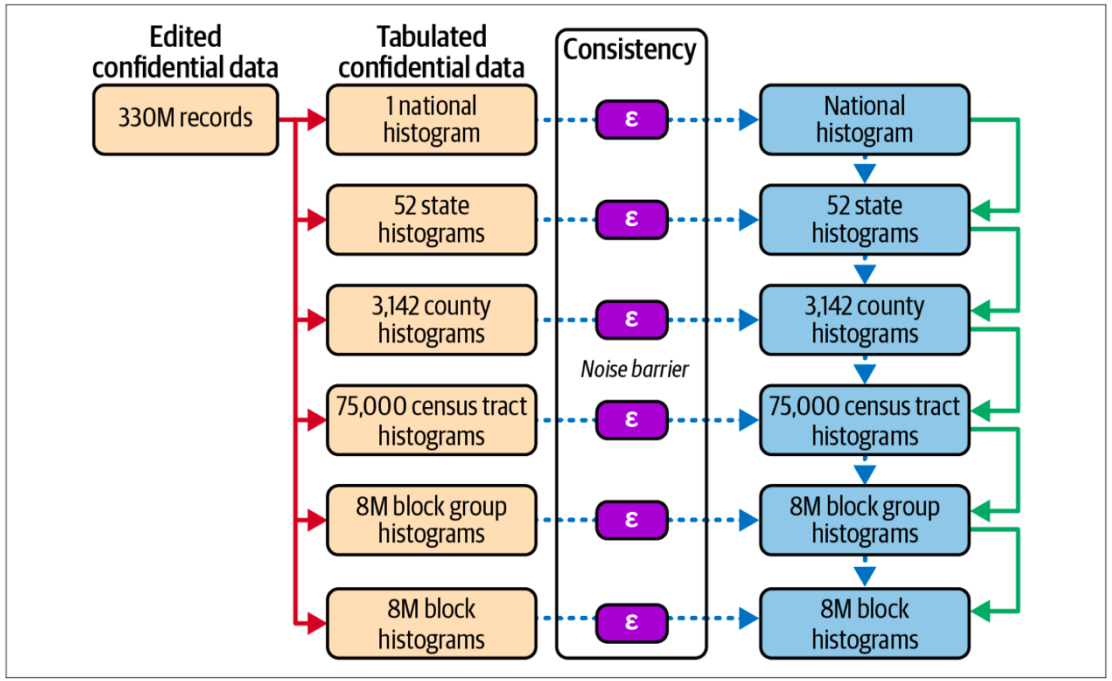
Anonymization in Data Security: Case Study of the US Census

Differential Privacy: The Solution

- Adds random noise to data
- Protects individuals while preserving patterns
- Balances privacy vs. data utility

The Implementation: Top-Down Approach

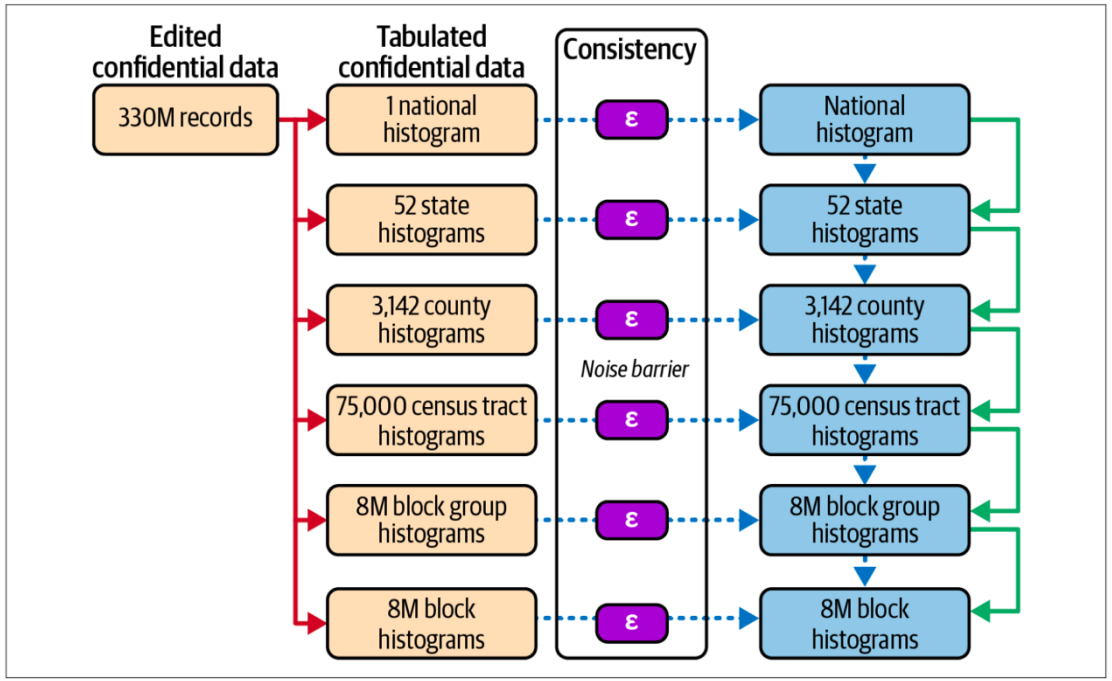
- Start from national level
- Add noise at each level: state → county → block
- Same overall statistics preserved



Anonymization in Data Security: Case Study of the US Census

The Challenge for Underrepresented Groups

- Risk of undercounting small populations
- Less representation, less funding
- Trade-off between privacy and equity



Ethical Considerations

- Underrepresented groups face greater privacy risks
- Often lack access to privacy tools
- Need to balance protection with fairness

Differential Privacy with the Laplace Mechanism

- Add noise to query results to protect individual privacy
- Goal: Add minimal noise while meeting privacy requirements
- Laplace distribution is commonly used to add this noise

$$\frac{1}{2b}\exp\left(-\frac{|x-\mu|}{b}\right)$$

μ is the location where the exponential functions meet (essentially the peak of the distribution)

b represents the scale of the distribution.

Sampling Noise from Laplace

- Noise sampled from Laplace distribution meets epsilon guarantees
- Looks like a double exponential—centered, sharp peak
- Useful because it often gives small noise, retaining information

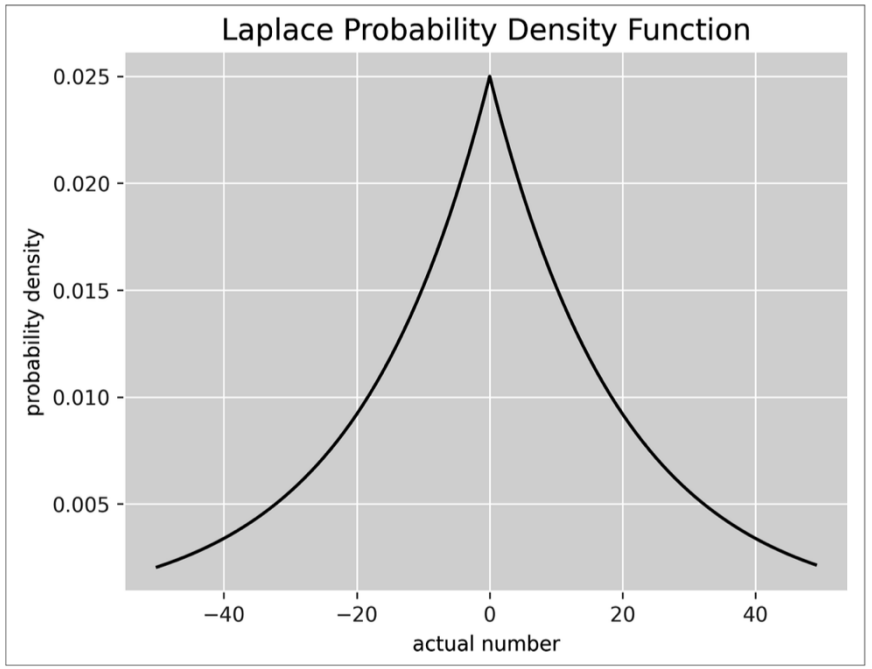


Figure 2-4. Laplace distribution

Differential Privacy with the Laplace Mechanism

Why Laplace Works

- High probability of small noise, but some chance of large noise
- Keeps most results close to original, enabling analysis
- Ensures plausible deniability: can't tell which data caused the noise

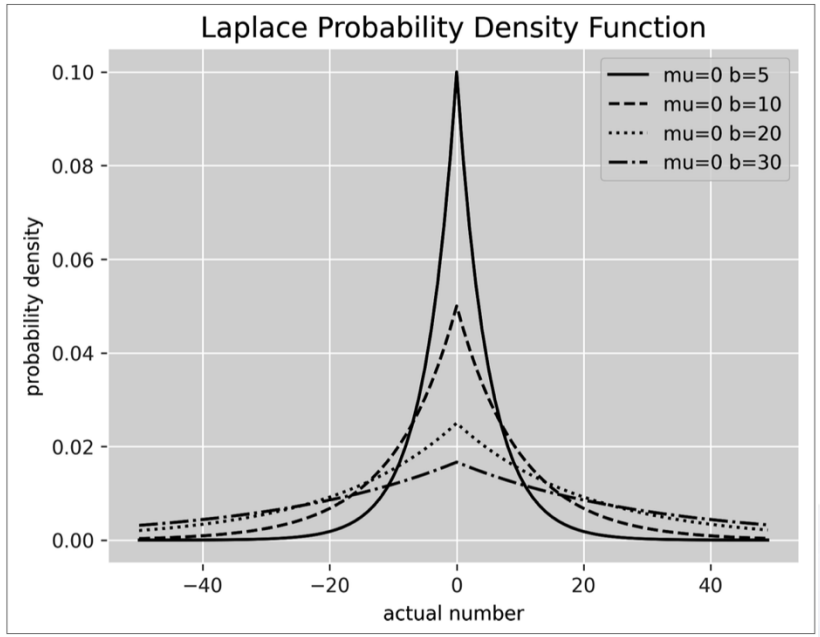


Figure 2-5. Laplace distribution: comparing scale and location

Differential Privacy with Laplace: A Naive Attempt

```
epsilon = 1.1

def naive_laplace_dp_mechanism(value, epsilon):
    # Please do not use this function, ever :)
    orig_value = value
    value =  np.random.laplace(value, 1/epsilon) # more on this choice later!
    print("Noise: {}".format(value - orig_value))
    return value

laplace_dp_mechanism(np.mean(ages), epsilon)
```

```
Noise: 1.277692233946226

46.55769223394623
```

Understanding Sensitivity

- Sensitivity = max change in query result or output from adding/removing one person
- Determines how much noise to add
- Example: Counting people adds 1; summing salaries can add much more

```
def filter_bounds(value, lower_bound, upper_bound):
    if value < lower_bound:
        return lower_bound
    elif value > upper_bound:
        return upper_bound
    return value

bounded_ages = [filter_bounds(age, 20, 70) for age in ages]
```

Bounding Sensitivity

- Use upper/lower bounds to limit query sensitivity
- Example: Limit age between 20–70 to cap sensitivity
- Larger bounds = more noise, smaller = less noise

Calculating with Bounded Data

- Use filter_bounds() to clip values to chosen limits
- Must apply to new users as they join the dataset
- This ensures consistent privacy guarantees across time

Privacy Budgets & Epsilon Composition

- Each query has its own epsilon (privacy cost)
- Total privacy loss = sum of epsilons for all queries
- Budgeting ε helps manage info gain vs privacy risk

For example:

I can budget my total epsilon appropriately over a certain number of queries.

Privacy Budgets & Epsilon Composition

Sensitivity in Real Queries

- Split ε between multiple queries: e.g. average = sum + count
- Use sensitivity/epsilon as the scale for Laplace noise
- Higher sensitivity → bigger scale → more noise

```
def laplace_dp_mechanism(value, epsilon, sensitivity):
    # Please do not use this function, ever :)
    orig_value = value
    value =  np.random.laplace(value, sensitivity/epsilon)
                                    # now you see why 1 was a poor choice!
    print("Noise: {}".format(value - orig_value))
    return value


epsilon_for_sum = 0.5
epsilon_for_count = 0.5
summed_ages = laplace_dp_mechanism(np.sum(bounded_ages),
                                    epsilon_for_sum, sensitivity=50)
count_ages = laplace_dp_mechanism(len(bounded_ages),
                                    epsilon_for_count, sensitivity=1)
summed_ages / count_ages
```

```
Noise: -53.24175804583501
Noise: 1.928454653399541

43.88135047434459
```

The actual mean = 45.28
Relative error -0.03

Exploring Gaussian Noise for Differential Privacy

- Alternative to Laplace: uses normal (Gaussian) distribution
- Still provides differential privacy guarantees
- Requires a relaxed definition → introduces delta (δ)

$$\Pr[A(D_1) \in S] \leq e^{\varepsilon} \cdot \Pr[A(D_2) \in S] + \delta$$

What is Delta (δ)?

- Delta allows a small probability of failure in privacy guarantee
- Enables release of a few rows without noise under tight bounds
- Delta should be cryptographically small (e.g., $< 1/n$)

Gaussian Mechanism Formula

- Gaussian mechanism is (ε, δ)-differentially private
- Only applies when $\varepsilon < 1$
- Adds more noise than Laplace
- Formula:

$$F(x) = f(x) + N(\sigma^2)$$

Where $\sigma^2 = \frac{2s^2 \log\left(\frac{1.25}{\delta}\right)}{\varepsilon^2}$, s = sensitivity

Comparing Laplace and Gaussian Noise

- Laplace: best when a person affects one value
- Gaussian: better if a person affects many values
- Also considers how often and how much someone contributes

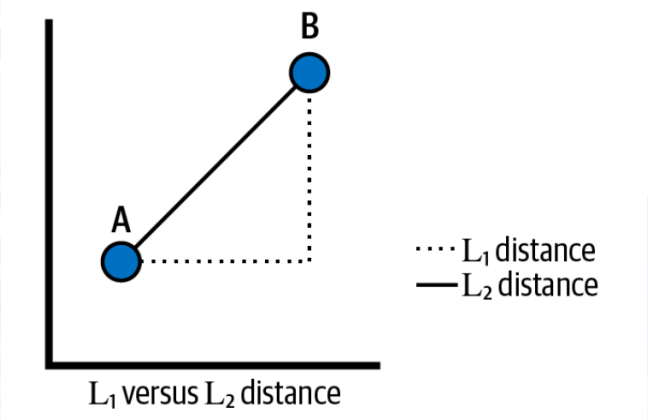
Clamping and Bounding for Gaussian Noise

- Use clamping bounds (e.g., age: 20–70, complaints: 0–10)
- Prevents extreme values from increasing sensitivity
- Bounded input = better control over noise

Comparing Laplace and Gaussian Noise

Distance Metrics – L_1 vs L_2

- Laplace uses L_1 (Manhattan distance)→ $\|x\|_1 = \sum|x_i|$
- Gaussian uses L_2 (Euclidean distance) → $\|x\|_2 = \sqrt{\sum|x_i|^2} = \sqrt{x^T x}$
- Affects how noise is added across multiple queries



Practical Insight – Noise Scaling

- Laplace noise scales linearly with affected queries (∝ k)
- Gaussian noise scales with $\sqrt{k}$
- Gaussian is more efficient when sensitivity is high

Implementing Gaussian Mechanism (Naive)

```
count_epsilon = 0.20
sum_epsilon = 0.79
delta=10**-5


def gaussian_dp_mechanism(value, epsilon, sensitivity=sensitivity):
    # Please do not use this function in real life - it is susceptible
    # to well-known attacks. Instead, use a well-known and audited
    # open-source differential privacy library
    orig_value = value
    gauss_scale = sqrt((2*sensitivity**2*log(1.25/delta))/epsilon**2)
    value =  np.random.normal(value, gauss_scale)
    print("Noise: {}".format(value - orig_value))
    return value

summed_ages = gaussian_dp_mechanism(np.sum(bounded_ages),
                                    sum_epsilon, sensitivity=50)
count_ages = gaussian_dp_mechanism(len(bounded_ages),
                                    count_epsilon, sensitivity=1)

mean_age = summed_ages / count_ages
mean_age
```

```
Noise: 332.0995166206894
Noise: 25.527328637215277

38.7015287377063
```

The actual mean = 45.28
Relative error 0.15 (5 times the Laplace re but using a smaller epsilon)

Debiasing Noisy Results

- Differential privacy adds noise → higher error
- Use denoising techniques (Bayesian, ML, priors)
- Gaussian works well: normal error is assumed in science

Sensitivity and Privacy Units

- Sensitivity typically means: 1 person added or removed
- This is known as a privacy unit
- But some cases (e.g., shared devices) need a broader definition

Long-Term Tracking & Budget Issues

- Repeated queries over time (e.g., location or browsing history) can:
  - Exhaust the privacy budget
  - Reveal more than intended
- Solution: bind the budget to a time interval (e.g., daily, weekly)

Risk of Small Privacy Units

- Smaller privacy units (per click, per day) can weaken protection
- Attackers with auxiliary info may connect repeated releases
- Result: users can be overexposed

Legal & Practical Guidance

- For GDPR or real-world compliance:
  - Collaborate with legal experts
  - Study literature on algorithms that match your goals
- Seek balance between utility and privacy

k-Anonymity

- Older anonymization approach (e.g., age, ZIP code, gender)
- Proposed by Latanya Sweeney to protect medical records
- Vulnerable to linkage attacks via voter data

K-Anonymity in Practice

Create groups where each contains $\geq k$ people

Table 2-2. Start dataset

Age	Zip code	Diagnosis
54	11189	Covid-19
44	11188	Normal Flu
22	11187	Covid-19
25	11189	Covid-19
44	34555	Covid-19
72	34555	Covid-19
66	34556	Normal Flu
77	34555	Normal Flu
78	11189	Normal Flu
75	33444	Covid-19

Table 2-3. Dataset with k=2

Age	Zip code	Diagnosis
40-60	1118*	Covid-19
40-60	1118*	Normal Flu
20-40	1118*	Covid-19
20-40	1118*	Covid-19
60+	3455*	Covid-19
60+	3455*	Normal Flu
60+	3455*	Normal Flu

- Remove "singletons" (unique individuals)
- No one can be singled out
- Still possible to infer sensitive info from majority values
- *E.g., all 20–40-year-olds in ZIP 1118 had COVID-19 → easy to guess*

K-Anonymity in Practice

Create groups where each contains $\geq k$ people

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Age	Zip code	Diagnosis
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Table 2-4. Dataset with k=2 and l=2

Age	Zip code	Diagnosis
40-60	1118*	Covid-19
40-60	1118*	Normal Flu
60+	3455*	Covid-19
60+	3455*	Normal Flu
60+	3455*	Normal Flu

Why K-Anonymity Fails

- Leads to homogeneous groups
- Cannot handle variance in sensitive attributes
- Introduced l-diversity: require sensitive attributes to vary

K-Anonymity in Practice

Create groups where each contains $\geq k$ people

Table 2-2. Start dataset

Age	Zip code	Diagnosis
54	11189	Covid-19
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60+	3455*	Normal Flu
60+	3455*	Normal Flu

Problems with L-Diversity

- L-diverse groups may be **imbalanced or skewed**
- Skew = statistical bias, harms ML/inference
- Dataset no longer reflects reality

K-Anonymity in Practice

T-Closeness

- Adds more constraint: distribution of attributes must match population
- Example: COVID vs flu ratio should reflect global stats
- Often results in no usable data left

Table 2-5. Dataset with k=2, l=2, and t-closeness

Age	Zip code	Diagnosis
None	None	None

Why K-Anonymity Isn't Enough

K-anonymity:

- Doesn't quantify privacy risk
- Requires manual, dataset-specific tuning
- Can introduce bias

Differential privacy is preferred:

- It quantifies risk
- Scales well
- Works across datasets

SUMMARY

- You learned to **define privacy** in a rigorous, scientific way
- Explored differential privacy as a **solution for anonymization**
- Compared **different types of noise** (Laplace, Gaussian)
- Created and managed a **privacy budget**
- Understood how **differential privacy works** in practice
- Addressed sensitivity and the **privacy–utility trade-off**
- Evaluated older methods like **k-anonymity** and **t-closeness**

Thank You

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